INCH-POUND

MIL-C-24643/17C
14 March 1994
SUPERSEDING
MIL-C-24643/17B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSFSGU
(INCLUDING VARIATION LSFSGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required:

Construction (watertight with circuit integrity)

* Sizes 3 to 100 inclusive, extruded silicone rubber insulation 1/

First - Copper conductor, uncoated, (see table I for size)
Second - Extruded Silicone rubber insulation (see table I for diameter).
* Third - Glass braid providing 80% coverage.
* Fourth - Braid covering (see table I for diameter). Standard identification code applied by method 1 or method 3.

Sizes 23 to 400 inclusive, silicone rubber-glass tape insulation 1/

First - Copper conductor, uncoated, (see table I for size)
Second - Silicone rubber - glass tape insulation (see table I for diameter). Letter identification code applied by method 5.
* Third - The four conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Fourth - A binder.

* 1/ Size 23 through 100 may be glass tape or extruded silicone rubber insulation.

AMSC N/A FSC 6145
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TABLE I. Details.

| Military part no. M24643/17 1/ | Size 2/ | Conductor Size AWG | Type's Overall diameter (inches) | | | | Diameter over outer insulation Tape nom. (inches) | Diameter over insulation min avg (inches) | Diameter over braid covering max (inches) | Jacket thickness min avg (inches) | Conductor resistance (ohms) | Insulation resistance (megohms) | Cold bending mandrel diameter max (inches) | Accel. service load-ing (amps) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | LSTSGU | | LSTSGA | | | | | | | | | |
| | | | min | max | min | max | | | | | | | | |
| -01UN | 3 | 16 (class B) | | 0.447 | | 0.497 | ---- | 0.096 | 0.130 | 0.030 | 4.3 | 500 | ---- | ---- |
| -02UN | 4 | 14 (class B) | | .513 | | .563 | ---- | .112 | .143 | .040 | 2.68 | 500 | ---- | ---- |
| -03UN | 9 | 10 (class B) | | .630 | | .680 | ---- | .154 | .187 | .040 | 1.08 | 500 | ---- | ---- |
| -04UN | 23 | 7 (class B) | | .890 | | .940 | 0.274 | .271 | .310 | .050 | .543 | 500 | ---- | 115 |
| -05UN | 50 | 3 (class C) | | 1.050 | | 1.100 | .334 | ---- | ---- | .050 | .21 | 200 | 13 | 185 |
| -06UN | 75 | 1 (class C) | | 1.240 | | 1.290 | .407 | ---- | ---- | .050 | .134 | 175 | 15 | 250 |
| -07UN | 100 | 0 (class D) | | 1.358 | | 1.408 | .453 | ---- | ---- | .060 | .106 | 160 | 16 | 295 |
| -08UN | 150 | 000 (class D) | | 1.625 | | 1.675 | .557 | ---- | ---- | .060 | .0674 | 135 | 19 | 400 |
| -09UN | 200 | 0000 (class D) | | 1.820 | | 1.870 | .634 | ---- | ---- | .060 | .0536 | 125 | 21 | 465 |

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.

2/ Type LSFSGU and LSFSGA





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Construction (watertight with circuit integrity) (continued)

Construction variant type LSFSGU

Sizes 3 to 100 inclusive, extruded silicone rubber insulation 1/

* Fifth - The four conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly
Sixth - Binder or combination binder and barrier.
Seventh - Cross-linked polyolefin jacket. (see table I for
* thickness).

Sizes 23 to 200 inclusive, silicone rubber-glass tape insulation 1/

Fifth - Cross-linked polyolefin jacket. (see table I for
* thickness).

Construction variant type LSFSGA

* Fifth - Cross-linked polyolefin jacket. (see table I for thickness).
* Sixth - Braided metal armor.

* 1/ Size 23 through 100 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| Basic electrical: | |
| Conductor Resistance - ohms/1000 feet at 25°C, maximum | (see table I) |
| Voltage withstand - volts, root mean square, minimum | |
| Sizes 3 through 9 | 3000 |
| Sizes 23 and larger | 5000 |
| Insulation resistance - megohms/1000 feet, minimum | (see table I) |
| Conductor continuity | No failure |
| Group A: | |
| Visual and dimensional | No failure |
| Watertightness - see MIL-C-24643 for limits of water leakage | No failure |
| Abrasion resistance - sizes 3 through 23 (extruded silicone rubber) scrapes, minimum | 250 |
| Crack resistance - sizes 3 through 23 (extruded silicone rubber) | No damage |



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EXAMINATION AND TESTS (continued):

| Group B: | Requirements |
|---|---|
| * Cross-linked proof test (percent, maximum) | |
| Insulation | 50 |
| Jacket (When tested at 200°C ) | 50 |
| Gas Flame - 1 hour | No failure |
| Drip - 95 ± 1°C | Zero |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Cold bending, cable - see table I for mandrel diameter | No damage |
| Physicals (unaged) | |
| Insulation (extruded) | |
| Tensile strength - lb/in$^2$, minimum | 700 |
| Elongation - percent, minimum | 250 |
| Jacket (cable) | |
| Tensile strength - lb/in$^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| Group C: | |
| Physicals (aged) | |
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged minimum | 60 |
| Hot oil immersion | |
| * Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Shrinkage | No failure |
| Heat distortion - percent of unaged, maximum | 30 |
| * Permanence of printing (conductor) - cycles, minimum (extruded silicon rubber only) | 25 |
| Permanence of printing (jacket) - cycles, minimum | 125 |
| * Cable sealant removability | No failure |
| Armor (type LSTSGA only) - conformance to material, construction and coverage | No failure |
| Group D: | |
| Flame propagation (cable) | No failure |

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:



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| QUALIFICATION INSPECTION: | Requirements |
|---|---|
| Cold working (minus 20 ± 2°C) | No damage |
| Gas flame (3 hours) | No failure |
| Specific gravity of extruded silicone insulation, maximum | 1.55 |
| Accelerated service - sizes 14 and larger (see table I for load current) | No failure |
| Aging and compatibility (cable) 125 ± 5°C) | No failure |
| Abrasion resistance (jacket) - scrapes, minimum | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| * Fillers | 45 |
| * Insulation | 35 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |

UNIT ORDERING LENGTHS:

| Type and Size | Feet (nominal) |
|---|---|
| LSFSGU - 3 | 2500 |
| LSFSGU - 4 and 9 | 2000 |
| LSFSGU - 23 and 50 | 1500 |
| LSFSGU - 75 through 150 | 1000 |
| LSFSGU - 200 | 500 |

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

(Project No. 6145-2057-17)



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